

Let's learn about **satellites!**

Space mission and **satellite** design course

Attitude Determination and Control System

What is satellite attitude?

We have already seen how to position a satellite in space to make it perform the required task: we discussed about the various types of orbit and what are they used for.

But it is not sufficient for a satellite to be in the right orbit: it also needs to be in the **right orientation**.

What happens if we need to take pictures of the Earth but the camera is pointing toward the Moon?

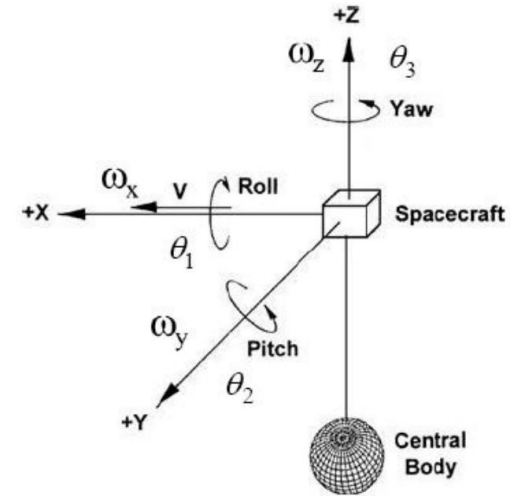
What happens if we need to send payload data to the ground station but the antenna is pointing toward the deep space?

And what if the solar panels are never facing the Sun?

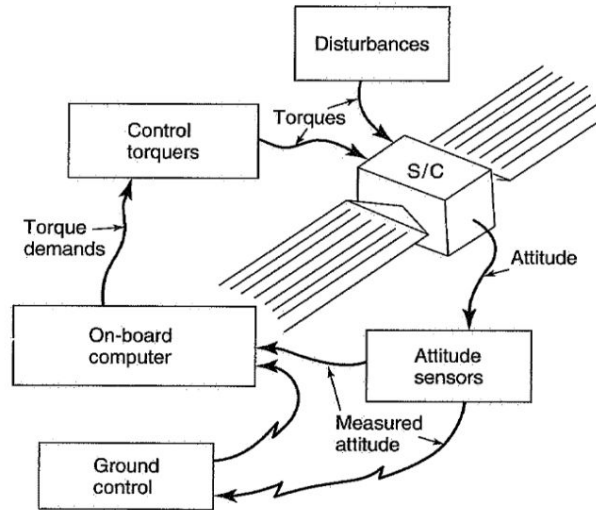
ADCS

The ADCS has two functions:

1. To **determine** satellite attitude using its **sensors**;
2. Control the attitude in order to **counteract** disturbance torques and to **maneuver** in order to meet the mission requirements, using its **actuators**.



Attitude Determination



The first step of an attitude determination and control sequence is always the determination of the attitude, and so to understand in which way the satellite is pointing, if it is rotating, at which angular velocity, and so on.

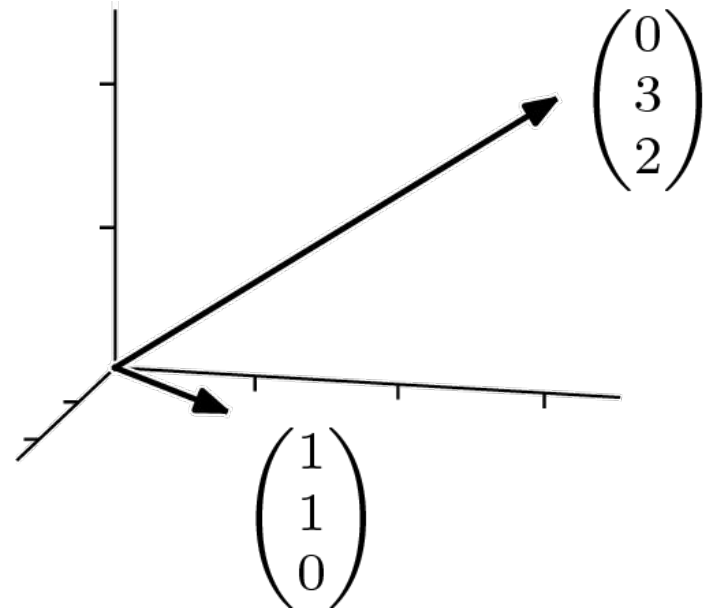
But, first of all, how do the attitude is described in numerical terms?

To answer these question first we need to talk about **attitude representation**.

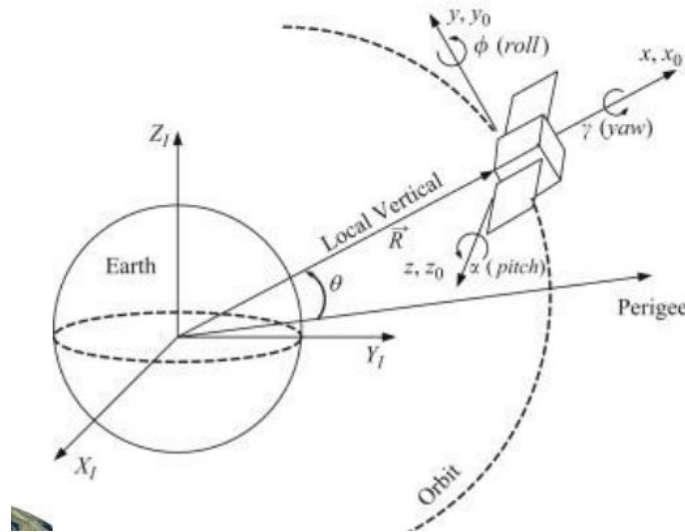
Vectors and versors

A vector is a set of scalar number that can have any dimension. If we use a vector of size 2, we can easily describe the position of a point in a plane. If we use a size 3 vector, we can describe the position of a point in a 3D space.

If the length of the vector (the black arrow in the figure) is equal to one, then the vector can be called **versor**.



Attitude Representation



The satellite has to be described in an Inertial Reference Frame.

We will use the **Earth Centered Inertial Frame**. A frame of reference can be described by 3 versor, in this case we call them X, Y and Z (or C1, C2 and C3).

Usually, the satellite attitude is described by the three angles **pitch**, **roll** and **yaw**, every one of them linked to one of the three satellite versors x,y z (or u, v w).

Attitude Determination Matrix

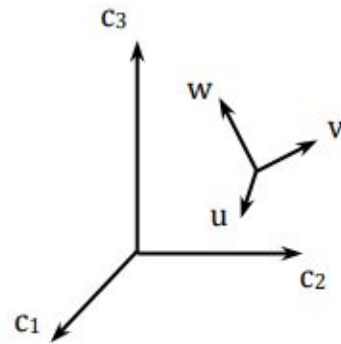
To accurately describe the satellite attitude, it is necessary to express the versors u, v, w in the reference frame $C1, C2, C3$. In this way you can build the **Attitude Matrix A**.

$$A = \begin{bmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{bmatrix}$$

$$\hat{u} = (u_1, u_2, u_3)^T$$

$$\hat{v} = (v_1, v_2, v_3)^T$$

$$\hat{w} = (w_1, w_2, w_3)^T$$



Attitude Determination

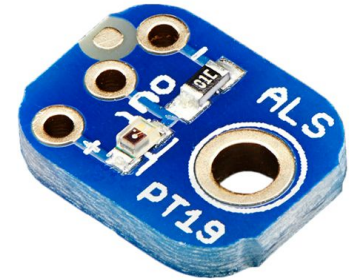
The attitude determination is the process of determining the matrix A , and so the relative orientation between the satellite and the reference frame of choice, in our case the ECI reference frame.

The attitude is determined using a series of **sensors**. They could rely on the observation of external references, like the Sun or the Earth, or with respect of internal references, like in the case of Inertial Measurement Unit.

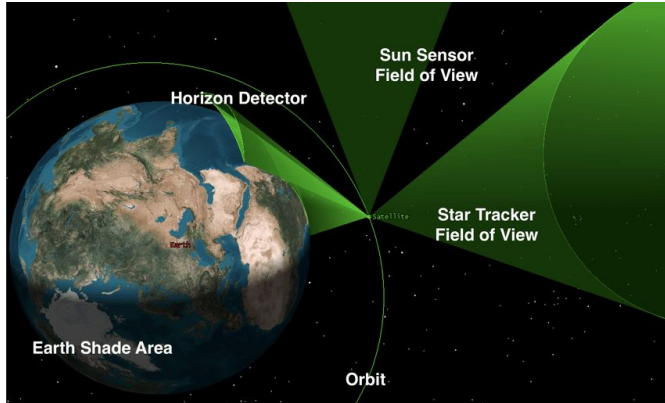
Sensors - 1

Sun Sensors: they measure the incidence angle of sunlight. They are simple and widely used, even if not so accurate and useless when the satellite is in eclipse. We have many of them in our education kit!

Star Sensors: they are much more complex and so more precise than Sun sensors. They are able to recognize one or more stars and to define the attitude based on their position.



Sensors - 2



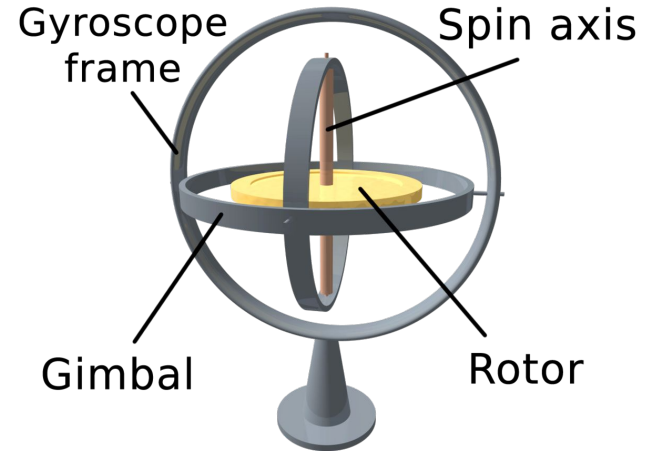
Horizon Detectors: they use an infrared sensor to detect the boundary between Earth atmosphere and deep space. They provide a clear direction pointing towards Earth, making them useful for nadir-pointing satellites.

Magnetometers: they can provide attitude information measuring the Earth magnetic field. Magnetometers are not very precise. Usually they're used with *Magnetorquers*.

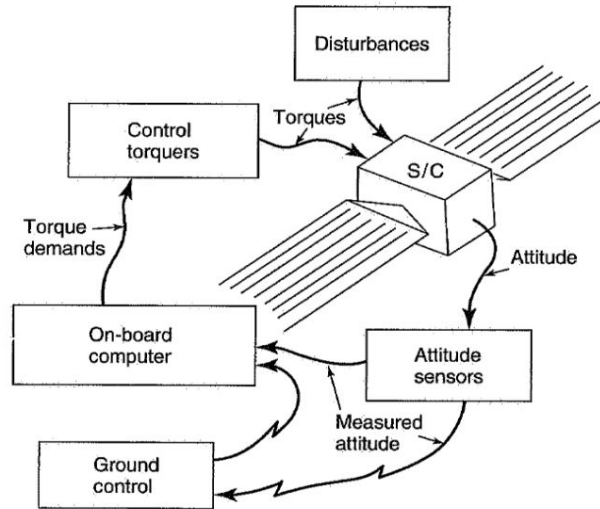
Sensors - 3

Gyroscopes or Inertial Measurement

Units: they are based on different technologies. They could resemble something like in the figure here, or they could be small electronic chips. Even in your phone you have IMU and gyroscopes! They measure acceleration, speed and position with respect to an internal reference. The error is accumulating over time though, so they need to be coupled with another sensor.



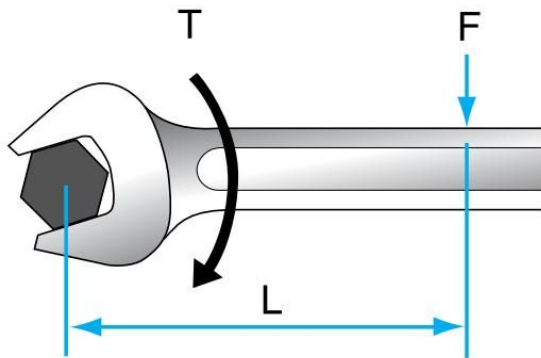
Attitude Control



Now that we have seen how to determine the attitude, we need to consider how to **control** it: how much force (or better, *torque*) we need to apply to the spacecraft? How do we apply it?

To continue, we need to define some physical principles like *torque*, *angular momentum* and *moment of inertia*.

What is torque?



$$\text{Torque } T = F (\text{Force}) \times L (\text{Length})$$

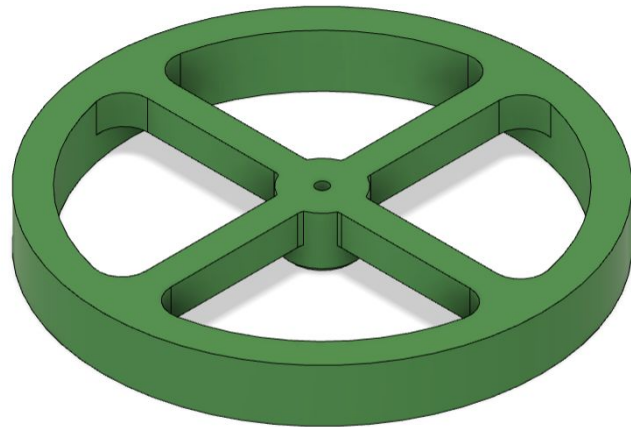
Torque is a force applied at a distance about a point, or fulcrum. Torque is so creating a *rotation* and not a linear movement.

Everytime you're turning something, you are applying some torque to it: think about fastening a screw with a screwdriver, or using a wrench, or even opening a plastic water bottle.

Moment of Inertia

A body is harder to turn if it has a higher moment of inertia. This is similar to the relation between mass and acceleration: the more mass a body has, the harder it is to push. The moment of Inertia is bigger if the mass is further away from the centre of rotation.

This is the reason why **Reaction and Momentum Wheels**, that need to have an high moment of inertia, are built with the mass located as further away from the centre as possible.



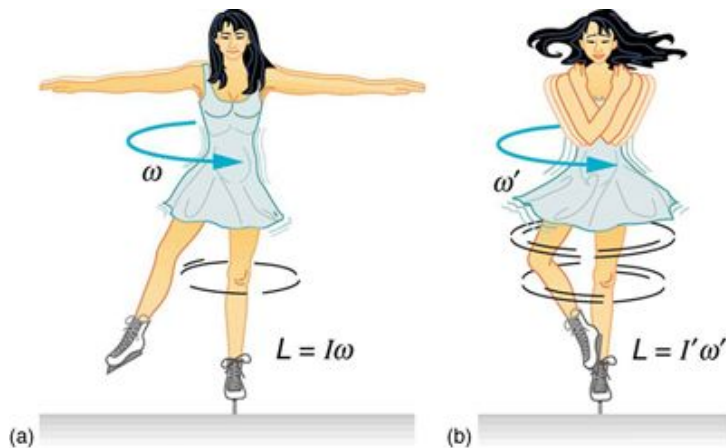
Angular Momentum

The product between inertia and angular velocity is called **Angular Momentum**.

$$L = I\omega$$

The rate of change of angular momentum of a body is equal to the Torque applied to the body.

Conservation of Angular Momentum



Angular momentum **is conserved** if there is no external torque. You can see this principle put in action by figure skaters: if they close their arms while they are spinning, their inertia decrease, since the mass is getting closer to the center. Since L is conserved, and thus remain equal between case a and case b, the angular velocity has to increase.

Satellites change their orientation in space in the exact same way, as we will see soon.

Disturbance Torques

We already know that we need to control the attitude in order to control the orientation of our satellite, but what happens once the satellite has the right orientation?

On orbit we have some **disturbance torques**, unwanted torques that the environment is applying on our satellite.

The disturbance torques are:

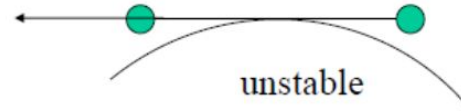
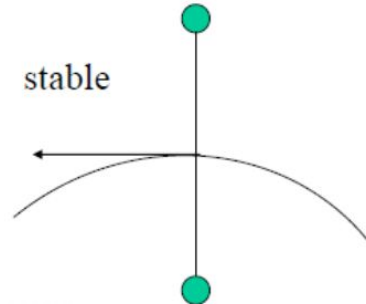
- **Gravity Gradient**
- **Solar Radiation Pressure**
- **Magnetic Field**
- **Aerodynamic Drag**

Gravity Gradient

If a satellite has an elongated shape, some configuration in orbits are more stable than other, as we can see in the figure; if our satellite need to operate by design in an unstable position, we need to counteract the gravity gradient torque. The maximum torque is given by:

$$T_g = \frac{3\mu}{2R^3} |I_z - I_y| \sin(2\theta)$$

where θ is the maximum deviation of z axis from the local vertical, I_z and I_y are the inertia about z and y .



Solar Radiation Pressure

The incident radiation of the Sun pushes against the satellite, as we have seen when we spoke about orbits. But what happens if the center of the solar pressure and the center of gravity of the satellite are not aligned? In this case, we have a force applied at a distance, and so a torque. The SRP torque is equal to:

$$T_{srp} = F(c_{srp} - c_g)$$

Magnetic Field

If the satellite has a residual dipole, the magnetic field of the Earth is applying a torque on the s/c.

This effect is particularly affected by the orbit shape, since the magnetic field of the Earth is different at different position in space. Furthermore, you want the residual dipole of the satellite to be as low as possible. The torque applied by Earth Magnetic Field is equal to:

$$T_m = DB$$

where D is the residual dipole of the spacecraft and B the Earth magnetic field.

Aerodynamic Drag

Similarly to what we have seen for SRP, if the center of aerodynamic pressure and the center of gravity of the satellite are not aligned we have a torque. Obviously, the lower is the orbit the greater is the effect of aerodynamic drag.

The aerodynamic torque is equal to:

$$T_a = F(c_{ap} - c_g)$$

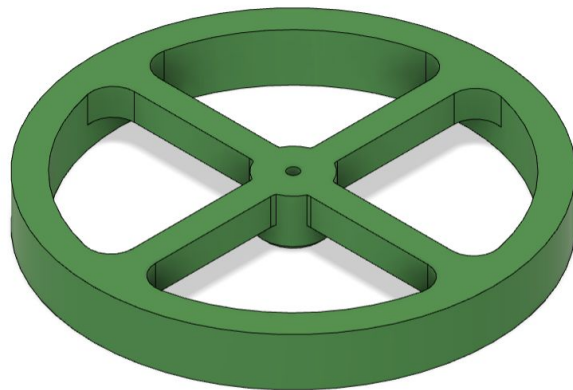
Actuators - 1

Now that we know what torques are affecting us, it is time to see what we can use to counteract them.

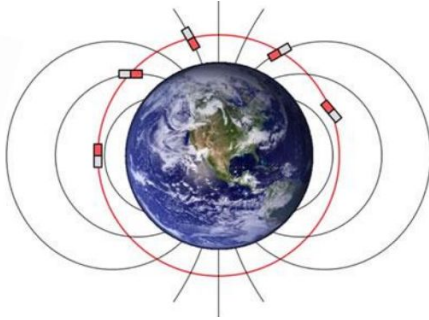
Reaction Wheels: heavy disks put in rotation by motors at varying velocity. The change in angular velocity create a torque to balance disturbances.

Momentum Wheels: similar to inertial wheels, but they are constantly spinning in order to achieve gyroscopic stiffness.

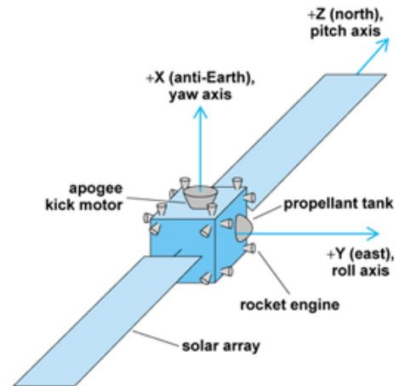
Control Moment Gyro: big and heavy spinning wheels; in this case the torque is applied tilting rotation axis. They are used only in big satellites.



Actuators - 2



Magnetic Torquers: electromagnets or coils that can create a torque proportional and perpendicular to Earth's magnetic field. We are using to our advantage one of the disturbances, by controlling the dipole D.



Thrusters: small rocket like elements are put in couple at a distance from the center of gravity, thus creating torques. In this case it is especially relevant the minimum quantity of thrust achievable by the system.

Passive Control - 1

Several control technique can be used in order to control the attitude of a satellite. They can be divided into **passive** methods, if they do not require power, or **active**. Let's start with an overview of the passive methods.

Gravity Gradient: useful for low altitude satellites; it makes use of the gravity gradient perturbation to remain in a stable configuration. Pointing accuracy is $\pm 5^\circ$.

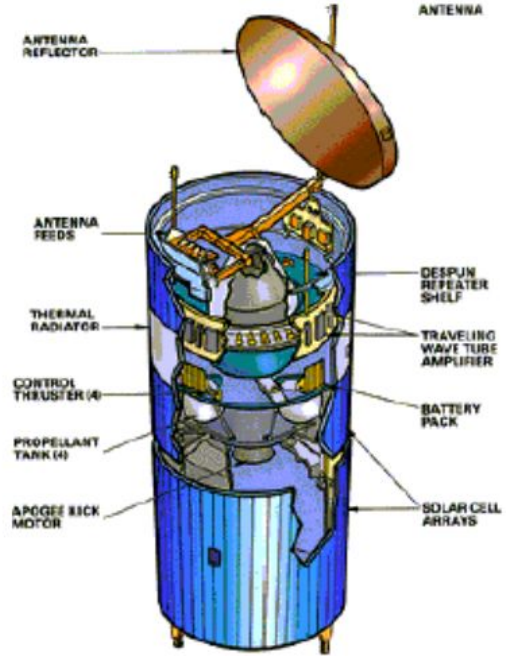
Magnetic Passive: useful for simple satellite in equatorial orbits; it makes use of permanent magnets on board that align the spacecraft with the Earth magnetic field. Even in this case, we have a pointing accuracy of $\pm 5^\circ$.

Passive Control - 2

Spin Stabilization: used for cylindrical satellites; the satellite is slowly spinning (20 to 90 rpm) gaining thus gyroscopic stiffness. It is also convenient for the natural scanning motion and for thermal control purposes.

The angular velocity and the correct spin axis have to be maintained by an active system, making so this approach more an hybrid type of control than a completely passive one.

The pointing accuracy ranges from $\pm 1^\circ$ to $\pm 0.1^\circ$



Active Control - 1

Otherwise, a satellite could be controlled fully on three axis, using the various actuators we described. Some common strategies are listed below.

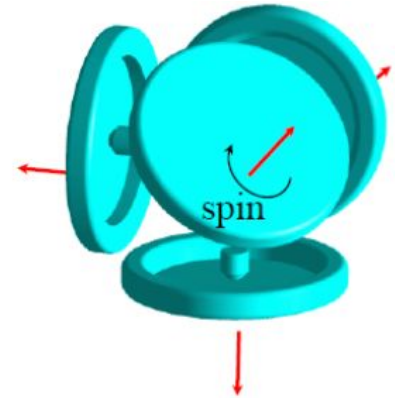
Momentum Bias: it consist in only one momentum wheel, that is rotating at high velocity. It is similar to the concept of a spin satellite, but in this case only the wheel is spinning. The pointing accuracy is $\pm 0.15^\circ$ to $\pm 0.05^\circ$.

Zero Momentum: three reaction wheels are counteracting the external torques varying their velocity, one for each of satellite axes. Usually, 4 wheel are used instead of three to increase the **redundancy** of the system. The pointing accuracy is $\pm 0.15^\circ$ to $\pm 0.05^\circ$.

Active Control - 2

The zero momentum configuration can be achieved also by other actuators, like thrusters, useful to counteract big torques, and control moment gyros.

Remember: every type of wheels needs to be desaturated. The wheels counteract torques increasing their speed; over time, the motor are no longer able to increase their speed, so they need another actuator to slow down and start the process again. So, wheel are always paired with other systems, usually with magnetorquers.



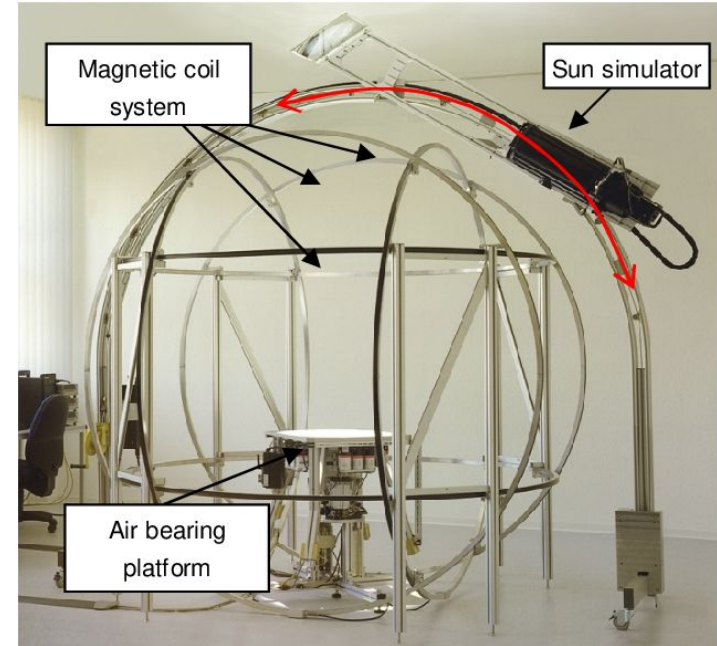
Active Control - 3



Attitude System Testing - 1

Once the ADCS is designed and the Control Method and algorithms are coded, it is necessary to **test** the correct functioning of the system.

There are several challenges in testing such a system on Earth; first of all, a spherical air bearing is responsible to generate a frictionless movement to simulate the orbit conditions. Then, the magnetic field of the earth is simulated by magnetic coils and a Sun simulator is put in place.



Attitude System Testing - 2



Mission Analysis - 1

Let's try to design together a ADCS subsystem. These kind of mission analyses are usually performed at a very early stage. First, the team decide what kind of control they need and perform some preliminary calculations; later on, when the satellite design is more refined, the ADCS team will make a more detailed design.

We want in this case to design an ADCS subsystem for a **Cubesat**, a small satellite (in this case, around 20x20x30 cm). The scope of the mission is to use a **optical payload** to obtain images of the **stars and distant galaxies**. The required pointing accuracy is around **0.1°**. The orbit will be at 550 km altitude.

Mission Analysis - 3

What kind of control technique is better to use?

- a. Passive Spin Stabilization
- b. Zero Momentum with thrusters
- c. Zero Momentum with Reaction Wheels and Magnetorquers
- d. Zero Momentum with Reaction Wheels

Mission Analysis - 6

What kind of sensors is better to use?

- a. Star Tracker
- b. Earth Sensor and Magnetometer
- c. Earth Sensor and Sun Sensor
- d. Star Tracker and Magnetometer

Mission Analysis - 8

Using the data shown below, compute all the disturbance torques in the **worst case scenario**, and the corresponding maximum external torque.

$$R_{\{Earth\}} = 6371 \text{ km}$$

$$h = 550 \text{ km}$$

$$c_{ap} - c_g = c_{srp} - c_g = 5 \text{ cm}$$

$$F_{drag} = 6.72 \times 10^{-3} \text{ N}$$

$$F_{srp} = 2.8 \times 10^{-4} \text{ N}$$

$$D = 0.05 \text{ Am}^2$$

$$B = 4.68 \times 10^{-5} \text{ T}$$

$$I_z = 1.63 \text{ kgm}^2$$

$$I_y = 1 \text{ kgm}^2$$

$$T_g = \frac{3\mu}{2R^3} |I_z - I_y| \sin(2\theta)$$

$$T_{srp} = F(c_{srp} - c_g)$$

$$T_m = DB$$

$$T_a = F(c_{ap} - c_g)$$

Mission Analysis - 9

Gravity Gradient Torque: remember that R is the Radius of the Earth + the orbit altitude and to convert the value in meters. Sinus of 2 times theta is considered to be 1, the maximum possible value, since we are in the worst case scenario.

$$T_g = 3.79 \times 10^{-7} Nm$$

Magnetic Field Torque: in this case, we only have to perform a multiplication.

$$T_m = 2.34 \times 10^{-6} Nm$$

Mission Analysis - 10

Solar Radiation Pressure and Aerodynamic Torques: even in this case we only have to perform a multiplication.

We obtain:

$$T_{\text{srp}} = 1.4 \times 10^{-5} \text{ Nm}$$

$$T_a = 3.36 \times 10^{-4} \text{ Nm}$$

Total External torque: we consider the worst possible case, so all of the external torques are added together.

We obtain:

$$T_{\text{ext}} = 3.53 \times 10^{-4} \text{ Nm}$$

Mission Analysis - 11

Considering the external torques, we have to choose which adcs to use. Here are listed 4 options with 5 different maximum torques. Which one do you choose for our satellite?

- a. $T = 3 \times 10^{-4} \text{ Nm}$
- b. $T = 3.53 \times 10^{-4} \text{ Nm}$
- c. $T = 5.3 \times 10^{-4} \text{ Nm}$
- d. $T = 4 \times 10^{-1} \text{ Nm}$